The *via negativa*: A general theory of structure

Abstract: We propose a general theory of structure, in which a structure, defined by a triple of domains, relations, and constraints, is instantiated iff its set of constraints is satisfiable. Our theory shifts focus away from structures under normal conditions to the conditions under which they might fail to be instantiated. Through our *via negativa* approach, we demonstrate how instances of theory change – the transition from naive to ZFC set theory, real to complex numbers, unrestricted to σ-additive measure in the σ-algebra, and so on – can be understood as instances of substitutive or subtractive soft lifts. This yields a unified treatment of classical paradoxes and impossibility results, including Russell's paradox, the non-existence of the largest natural number, $x^2 = -1$ over $\mathbb{R}$, Penrose-style geometries, and non-measurable sets. In addition, instead of taking levels as ontologically primitive, we derive a hierarchy of levels from these lifts. Through intra-level lifts, we can define each level as an equivalence class of structures. Inter-level lifts yield an ordered relation over levels. Higher-level constraints decompose into reducible and autonomous components, providing a criterion for emergence. Our resulting framework is a unified account of structure, emergence, and a level hierarchy, with instances of structural failure illuminating how obstructions result and might be overcome across logic, mathematics, and physics.

Keywords: Structural realism; structural failure; paradoxes; impossibility results; level; emergence; lift

§1 Intellectual background and case for novelty

Is there a mind-independent world? According to metaphysical idealism, nothing exists except minds and ideas in minds and there is therefore no mind-independent world (Berkeley, 1710). Our position is a metaphysical realist one: we assert the existence of an objective, mind-independent world (Aristotle, 1924; Locke, 1690; Moore, 1939). In addition, our position is an ontic structuralist one, since we believe that objects in this mind-independent world can be reconceptualized as structures (Costa and French, 2003; French, 1999, 2014; French and Ladyman, 2003a,b; Ladyman and Ross, 2007).

Our position draws on a broad, expansive tradition of structural realism (Carnap, 1967; Poincaré, 1898; Russell, 1927; Worrall, 1989). It is also informed by the debate in the philosophy of mathematics concerning whether mathematical structures exist independently (*ante rem* structuralism), only in their instances (*in re* structuralism), or as structural possibilities (modal structuralism) (Benacerraf, 1965; Hellman, 1989; Resnik, 1997; Shapiro, 1997, 2008). It is equally influenced by the debate in the philosophy of levels and emergence concerning whether emergent levels are real or metaphysically mysterious, reducible to physics or autonomous, and explainable in terms of mechanistic, grounding, or dependence relations (Anderson, 1972; Craver, 2007, 2013, 2014; Fodor, 1997; Fodor, 1974; Wilson, 2013, 2021). In addition, the formal tools that support our analysis hail from model-theoretic semantics (satisfiability), category theory (lifts and morphisms), and the logic of constraint satisfaction (instantiation as the simultaneous satisfaction of a set of constraints) (Awodey, 2010; Hodges, 2001; Mackworth, 1977, 1992; MacLane, 1971; Tarski and Woodger, 1983).

Against this intellectual backdrop, the following would comprise our claims to novelty:

> (Method) Our theory of structure makes instances of structural failure the theoretical load-bearer. Our *via negativa* approach is motivated in turn by two assumptions:
> (A1) Structures under normal conditions are overdetermined;
> (A2) Whereas normal operation hides structure, instances of structural failure (pathological) allow us to learn more about these structures.
> This *via negativa* approach has analogues in lesion studies (pathological) in neuroscience; knockout-gene studies (pathological) in model organisms; the investigation of material or structural failure or malfunction (pathological) in forensic engineering; and the removal of AI components (pathological) in ablation studies;
> (Novelty relative to structural realists) Structural realists typically treat structure under normal conditions as primitive and then proceed to worry about the nature of the object/structure distinction.[1] As a result, they have no account of structural failure: failed structures are simply absent from their account. By contrast, as our theory of structure makes instances of structural

[1] Critics of structural realism tend to make a similar move. For instance, see Psillos (1995, 2006).

failure the theoretical load-bearer, failed structures are informative;

(Novelty relative to the philosophy of levels and emergence) The standard literature here typically assumes levels as ontologically primitive. By contrast, our theory of structure demonstrates how levels are derived rather than stipulated by fiat. From our analysus, levels emerge as equivalence classes under intra-level lifts, while the ordered relation of a level hierarchy results from inter-level lifts. Emergence and reduction, in turn, become structural properties that can be inferred from our ordered relation;

(Novelty relative to model theory) Whereas model theorists typically study satisfiable structures under a model, our theory of structure systematically investigates unsatisfiable structures and how they might get repaired;

(Novelty relative to category theory) Whereas category theorists typically study morphisms in the abstract sense, our theory of structure systematically connects morphisms to constraint satisfaction, theory and model change, levels, a level hierarchy, emergence, and reduction;

(Novel synthesis under a single explanatory framework) The diverse intellectual influences that inform our position have been organized and integrated into a unified explanatory framework. This framework allows us to make philosophically important distinctions (intra- and inter-level lifts, subtractive and substitutive soft lifts, and so on); treat classical paradoxes and impossibility results as data rather than anomalies; understand theory and model change in terms of the engineering of constraints (soft lifts) to meet the demands of satisfiability; derive levels (equivalence classes of structures), hierarchies (an ordered relation on levels induced by lifts), and emergence (the appearance of autonomous, irreducible constraints when higher-level constraints are decomposed); and integrate physics in a precise fashion (physics as mathematical instantiability filtered by nomological constraints rather than mere applied mathematics).

§2 Sketch of our theory

Our theory of structure involves making sense of both structural instantiation and structural failure in terms of domains (Δ), relations ($\mathcal{R}$), and a hierarchy of constraints (Φ). Δ fixes the space of possible structures, $\mathcal{R}$ identifies specific structures in that space in terms of relational patterns, and Φ selects which of these specified structures are admissible for instantiation. A structure $\mathcal{M}$ can be defined in terms of the triple (Δ, $\mathcal{R}$, Φ).[2] $\mathcal{M}$ is instantiated iff (Δ, $\mathcal{R}$, Φ) is satisfiable. Formally:

$$\exists \mathcal{M}(\mathcal{M} \models \Phi)$$

By contrast, a structure fails or cannot be instantiated when (Δ, $\mathcal{R}$, Φ) is unsatisfiable. Formally:

$$\nexists \mathcal{M}(\mathcal{M} \models \Phi)$$

No level is presupposed as an ontological primitive. A projection $\pi : \mathcal{M}_2 \rightrightarrows \mathcal{M}_1$ from a richer source structure ($\mathcal{M}_2$) to a simpler target structure

[2] Alternatively, we could define bare structure $\mathcal{M}_0$ in terms of the 2-ple (Δ, $\mathcal{R}$) and treat Φ as instantiating constraints. Relative to this skeleton alternative, $\mathcal{M}_0$ is instantiated iff (Δ, $\mathcal{R}$, Φ) is satisfiable.

($\mathcal{M}_1$) is a triple of maps: it is surjective on domains ($\pi_\Delta : \Delta_2 \rightrightarrows \Delta_1$), maps relation-types to relation-types ($\pi_\mathcal{R} : \mathcal{R}_2 \rightrightarrows \mathcal{R}_1$), and constraints to constraints ($\pi_\Phi : \Phi_2 \rightrightarrows \Phi_1$). A morphism $\iota : \mathcal{M}_1 \hookrightarrow_D \mathcal{M}_2$ is an intra-level lift, just in case it satisfies the following conditions:

1. Domain expansion: $\Delta_1 \subseteq \Delta_2$;
2. Relation-type preservation: $\mathcal{R}_2 \supseteq \mathcal{R}_1$ The relation-type signature remains invariant across structures $\mathcal{M}_1$ and $\mathcal{M}_2$;
3. Constraint disposition: Two constraint disposition regimes, determined by the satisfiability of Φ_1 on $(\Delta_2, \mathcal{R}_2)$:
Strict lift: If Φ_1 is globally satisfiable on $(\Delta_2, \mathcal{R}_2)$, then $\Phi_2 = \Phi_1$ (all constraints are preserved);
Soft lift: If Φ_1 is not globally satisfiable on $(\Delta_2, \mathcal{R}_2)$, then $\Phi_2 \subsetneq \Phi_1$ (a maximal satisfiable subset of Φ_1 is retained and $(\Delta_2, \mathcal{R}_2, \Phi_2)$ is satisfiable;
4. Projection fidelity: There exists a projection $\pi_\emptyset : \mathcal{M}_2 \rightrightarrows_\emptyset \mathcal{M}_1$ that contracts the domain ($\Delta_2 \to \Delta_1$) and forgets none of the relation-types ('$\emptyset$' denotes that nothing in the relation-type signature is erased).[3]

A morphism $\lambda : \mathcal{M}_1 \uparrow \mathcal{M}_2$ is an inter-level lift, just in case it satisfies the following conditions:

1. Domain composition: There exists a surjective map $\gamma : \mathcal{P}(\Delta_1) \to \Delta_2$ that partitions Δ_1 into non-overlapping bundles, with Δ_1 and Δ_2 not in bijection;[4]

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[3] More than a book-keeping device, the projection $\pi_\emptyset$ functions as the structural witness, capable of 'testifying' that $\mathcal{M}_2$ is conservative over $\mathcal{M}_1$.

[4] This allows us to treat higher-level structures as bundles of lower-level constituents. As

2. Grounding correspondence: There exists a map $\rho : \mathcal{R}_2 \to \mathrm{Expr}(\mathcal{R}_1)$ that maps each upper relation-type to a predicate expression over $\mathcal{R}_1$, such that no $\nabla_2 \in \mathcal{R}_2$ (associated with richer, higher-level structure $\mathcal{M}_2$) is synonymous with any single $\nabla_1 \in \mathcal{R}_1$ (associated with simpler, lower-level structure $\mathcal{M}_1$);
3. Constraint reconstruction: The upper constraint set Φ_2 decomposes as $\Phi_2 \coloneqq \Phi_2^R \cap \Phi_2^A$, where Φ_2^R are constraints derivable from below (fully determined by the lower level and connecting relations between $\mathcal{M}_1$ and $\mathcal{M}_2$) and Φ_2^A are constraints autonomous from below (novel, emergent constraints that exist only at the higher level);
4. Dissolution fidelity: There exists a projection $\pi_{\downarrow}\mathcal{M}_2 \rightrightarrows \mathcal{M}_1$ that allows me to dissolve the higher-level structure $\mathcal{M}_2$ into its lower-level constituents. The residue $\mathcal{M}_1 \setminus \pi_{\downarrow}(\mathcal{M}_2)$ quantifies the underdetermination $\mathcal{M}_2$ relative to $\mathcal{M}_1$.

Two structures $\mathcal{M}_1$ and $\mathcal{M}_2$ are at the same level, just in case one of the following conditions holds:

1. Level identity ($\mathcal{M}_1 \cong_L \mathcal{M}_2$): They are instantiated in the same domain and under the same constraints. Formally, $\Delta_1 = \Delta_2, \Phi_1 = \Phi_2$, and ($\Delta_1$, $\mathcal{R}_1$, Φ_1) and (Δ_2, $\mathcal{R}_2$, Φ_2) are both satisfiable;
2. Level equivalence ($\mathcal{M}_1 \sim_L \mathcal{M}_2$): They are connected by an intra-level lift $\iota : \mathcal{M}_1 \hookrightarrow_D \mathcal{M}_2$.

A level L_i is an equivalence class of structures that either are instan-

γ encodes which L_1 stuff gets bundled together to constitute the emergent L_2 structure, it formalizes the emergence of new properties.

tiated under the same constraint-domain pairs (level identity or $\cong_L$) or are connected by a chain of intra-level lifts (level equivalence or $\sim_L$). Intra-level lifts imply that we can stay on the same level, even when we expand the domain or relax constraints. A partial order $\prec_L$ on levels or level hierarchy can be defined in terms of the inter-level lift $\lambda : \mathcal{M}_1 \uparrow \mathcal{M}_2$. Level L_1 is strictly below level L_2 ($L_1 \prec L_2$), just in case there exists an inter-level lift $\lambda : \mathcal{M}_1 \uparrow \mathcal{M}_2$ with $\mathcal{M}_1 \in L_1$ and $\mathcal{M}_2 \in L_2$ and no intra-level lift ι between any member of L_1 and any member of L_2. Although no level is presupposed as an ontological primitive, levels can be derived from the lift relations ι (intra-level) and λ (inter-level). Furthermore, emergence and reduction become structural properties that can be inferred from our ordered relation $\prec_L$. When an inter-level lift $\lambda : \mathcal{M}_1 \uparrow \mathcal{M}_2$ has $\Phi_2^A \neq \emptyset$, the transition from L_1 to L_2 is emergent. By contrast, when $\Phi_2^A = \emptyset$, L_2 can be fully reduced to L_1.

Our theory of structure implies a pipeline of constraints, with constraints at each level either being emergent or derivable from lower levels. The base level L_1, logic, is governed by logical constraints (φ_1). At L_2, mathematical constraints (φ_2) ensure embeddability into a mathematical interpretation and give rise to structure-preserving mathematical embeddings. At L_3, nomological constraints (φ_3) ensure satisfiability in a mathematical model consistent with the laws of physics and give rise to dynamically realizable physical trajectories.[5] For the rest of this paper, we shall focus on instances of structural failure and demonstrate how structural failure can

[5] While our analysis has been confined to logic, mathematics, and physics, we believe that higher-level constraints can be anticipated through an appropriate extension of our theory of structure. For instance, we anticipate that chemical constraints (φ_4) at L_4 ensure compositional stability under interaction and give rise to energetically stable chemical configurations. We equally anticipate that biological constraints (φ_5) at L_5 ensure membership in an evolutionary lineage comprising functionally self-organized individuals and give rise to evolutionarily connected biological organisms. Defending such an extension to our theory lies, however, beyond the current scope of our paper.

both provide support for and further enrich our theory of structure.

§3 Structural failure at φ_1

In the space of logically possible structures, at least some relational structures, though logically conceivable and syntactically well-defined, are logically inconsistent. Examples of logically conceivable though inconsistent structures include the Russell set and the largest natural number. The Russell set R refers to the set of all sets that are not members of themselves.

1. ass.: $\exists R(R \coloneqq \{x \mid x \notin x\})$ (definition of the Russell set R)
2. If $R \in R$, then $R \notin R$ – from 1
3. Equally, if $R \notin R$, then $R \in R$ – from 1
4. Therefore, $\nexists R$ (from 1, 2 contradicts 3)

This is known as Russell's paradox (Frege, 1964, 1967; Russell, 1967). Where Δ is the universe of sets, $\mathcal{R}$ is the set membership ($\in$) relation, and Φ_{naive} is the axiomatic schema of naive set theory, Russell's paradox shows that there is no structure R such that (Δ, $\mathcal{R}$, Φ_{naive}) is satisfiable. While the Russell set is logically conceivable, it is fails to be instantiated because no model of Φ_{naive} exists anywhere. In particular, Φ_{naive} contains the unrestricted comprehension principle:

$$\text{(UCP)}\ \forall F\, \exists S\, \forall x\, \big(x \in S \ \leftrightarrow\ F(x)\big)$$

The Russell set R or set of all sets that are not members of themselves, though impossible, satisfies $F(x) = x \notin x$. Removing UCP from Φ_{naive} and replacing it with a restricted comprehension principle ($\Phi_{\text{restricted}}$) yields the axiomatic schema of ZFC set theory or Zermelo–Fraenkel set theory with the axiom of choice (Φ_{ZFC}). Φ_{ZFC} can be defined as $\Phi_{\mathrm{naive}} \setminus \mathrm{UCP} \cup \Phi_{\mathrm{restricted}}$. In addition, Φ_{ZFC} allows us to generate the von Neumann universe $\mathcal{V}$, a globally consistent domain of well-founded sets that excludes the Russell set R. We have already defined a soft lift in the following manner:

> Soft lift: If Φ_1 is not globally satisfiable on $(\Delta_2, \mathcal{R}_2)$, then $\Phi_2 \subsetneq \Phi_1$ (a maximal satisfiable subset of Φ_1 is retained and $(\Delta_2, \mathcal{R}_2, \Phi_2)$ is satisfiable. (§ 2)

We can now further distinguish between two types of soft lift:

> Subtractive soft lift: $\Phi_2 = \Phi_1 \setminus S$ for some unsatisfiable subset S of constraints;
> Substitutive soft lift: $\Phi_2 = (\Phi_1 \setminus S) \cup \Phi_{\mathrm{new}}$

Φ_{ZFC} can be classified as a substitutive soft lift from Φ_{naive}, with $\Phi_{\mathrm{restricted}}$ as the additional new constraint (Φ_{new}) that ensures expressive power without restoring the Russell paradox. Another instance of structural failure involves the notion of the largest natural number N. Here, Δ is the set of natural numbers $\mathbb{N}$, $\mathcal{R}$ is both the successor relation S and the ordering relation $<$, and Φ_{PA} refers to the constraints introduced by the Peano axioms. Two specific constraints within Φ_{PA} are relevant to this instance of structural failure:

> (Φ_S) Every natural number has a successor: $\forall x \exists y (x \in \mathbb{N} \to y = S(x))$;
> ($\Phi_<$) Every natural number is smaller than its successor: $\forall x (x < S(x))$

The assumption in favour of N adds a new constraint:

$(\Phi_{\max})\ \exists N \forall y (y \in \mathbb{N} \to y \leq N)$

However, no model exists in which all three constraints Φ_S, $\Phi_<$, and $\Phi_{\max}$ hold. This makes $\Phi_{\text{PA}} \cup \Phi_{\max}$ unsatisfiable:

1. ass. $\exists N$(N is the largest natural number)
2. Since $N \in \mathbb{N}$, N must have a successor $S(N)$ (from Φ_S)
3. $N > S(N)$ (from $\Phi_{\max}$)
4. $N < S(N)$ (from $\Phi_<$)
5. Therefore, $\nexists N$ (from 1, 3 contradicts 4)

N violates the asymmetry of $>$ on $\mathbb{N}$ in all models satisfying the Peano axioms.[6] Both the Russell set and the largest natural number are structural failures that arise from global inconsistency. However, unlike in the Russell set case, no additional constraint is required: we can simply reject $\Phi_{\max}$ (an external assumption inconsistent with Φ_{PA}), and this leaves $\mathbb{N}$ intact. This is tantamount to a subtractive soft lift.

§4 Structural failure at φ_2

Next, consider the algebraic equation $x^2 = -1$. $x^2 = -1$ is logically conceivable, since squaring is well-defined. Does $x^2 = -1$ have a solution? Δ is the set of real numbers $\mathbb{R}$, $\mathcal{R}$ consists of the arithmetic relations $\{+, \times, <\}$, and $\Phi_{\mathbb{R}}$ refers to constraints governing real numbers. Two constraints within $\Phi_{\mathbb{R}}$ warrant consideration:

[6] Indeed, the set of natural numbers $\mathbb{N}$ is open-ended and has no maximal element.

(Φ_{ord}) $\forall a(a \in \mathbb{R} \to (a^2 \geq 0)$;
(Φ_{neg}) -1 $<$ 0

$x^2 = -1$ introduces an additional constraint:

(Φ_i) $\exists i(i \in \mathbb{R} \ \wedge i^2 = -1)$

However, no model exists in which all three constraints Φ_{ord}, Φ_{neg}, and Φ_i hold. This makes $\Phi_{\mathbb{R}} \cup \Phi_i$ unsatisfiable:

1. ass. $\exists a \in \mathbb{R}$ such that $a^2 = -1$ (Φ_i)
2. $\forall a \in \mathbb{R}(a^2 \geq 0)$ (from Φ_{ord})
3. -1 $\geq$ 0 (from 1 and 2)
4. -1 $<$ 0 (from Φ_{neg})
5. Therefore, $\nexists a$ (from 1, 3 contradicts 4)

$x^2 = -1$ violates the asymmetry of $>$ on $\mathbb{R}$ and cannot be embedded into its intended structure in $\mathbb{R}$. Unlike the largest natural number case in § 3, however, the structural failure here is not terminal. An intra-level lift $\iota : \mathcal{M}_{\mathbb{R}} \hookrightarrow_{\Delta} \mathcal{M}_{\mathbb{C}}$ allows us to expand our domain to the domain of complex numbers $\mathbb{C}$, drop Φ_{ord}, and add a new constraint for algebraic closure:

(Φ_{alg}) $\forall p\Big((p \in \mathbb{C}[x] \wedge \deg(p) \geq 1) \to \exists z(z \in \mathbb{C} \wedge p(z) = 0)\Big)$

Where $\mathbb{C}$[x] denotes the set of all polynomials with complex coefficients, deg(p) denotes the degree of the polynomial, and $p(z) = 0$ denotes that z is the root of polynomial p, Φ_{alg} informs us that every non-constant polynomial with complex coefficients has at least one complex root.[7] The construction

[7] This is also known as the fundamental theorem of algebra (Argand, 1815; Gauss, 1799).

$\mathbb{C} \coloneqq \mathbb{R}[x]/(x^2+1)$ yields $\mathbb{C}$ as an algebraically closed domain, with Φ_{alg} in turn quantifying over $\mathbb{C}$. Since Φ_{ord} does not allow for x^2+1 to have any roots, it is dropped. Φ_{alg} is consistent with $\Phi_{\mathbb{R}} \setminus \Phi_{\text{ord}}$. Furthermore, as x^2+1 is a degree-2 polynomial in $\mathbb{C}$[x], Φ_{alg} instantiates i as a root. The intra-level lift $\iota : \mathcal{M}_{\mathbb{R}} \hookrightarrow_{\Delta} \mathcal{M}_{\mathbb{C}}$ can be classified as a substitutive soft lift, with Φ_{ord} being replaced with Φ_{alg}, allowing us to overcome an obstruction under $\mathbb{R}$ and instantiate a solution for $x^2 = -1$ under $\mathbb{C}$.

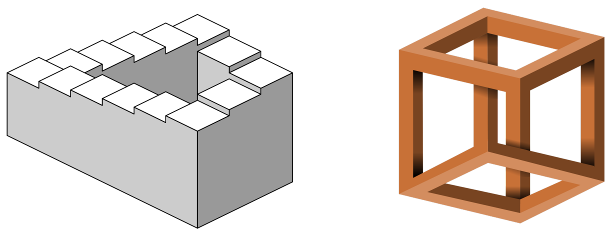

Figure 1: From left to right: the Penrose steps and the impossible cube, formalizable as a set of lines on a plane but logically incoherent under their intended interpretation in 3D Euclidean space

In a related vein, the Penrose steps and the impossible cube are logically conceivable: describing their local relationships – lines, angles, junctions – does not involve any straightforward contradiction (Fig. 1). Let S denote the set of steps making up the Penrose steps and let $\mathbb{R}$ denote the set of real numbers representing the space of possible heights for each step on the staircase. The function f embeds these steps in 3D Euclidean space ($\mathbb{R}^3$), while the function h assigns to each step a real-valued height. In addition, Φ_E denotes the axiomatic schema of 3D Euclidean geometry. The Penrose steps are characterized in terms of a pair of constraints:

(Φ_{loc}) There is local ascent at each step: $\forall i(h(x_{i+1}) > h(x_i))$;

(Φ_{loop}) The Penrose staircase is globally closed: $x_1 \to x_2 \to x_3 \to \cdots \to x_n \to x_1$

At the same time, there is one constraint within Φ_E that generates a contradiction:

($\Phi_{\text{s.t.}}$) $<$ is a strict total order on $\mathbb{R}$, such that:
($\Phi_{\text{s.t.}_1}$) It is irreflexive: $\neg(a < a)$;
($\Phi_{\text{s.t.}_2}$) It is asymmetrical: If $a < b$, then $\neg(b < a)$;
($\Phi_{\text{s.t.}_3}$) It is transitive: If $a < b$ and $b < c$, then $a < c$

Δ is the set of steps S, $\mathcal{R}$ consists of the functions and arithmetic relations $\{f, h, <\}$, and Φ_E refers to the Euclidean constraints governing the intended embedding under $\mathbb{R}^3$. No model satisfies Φ_{loc}, Φ_{loop}, and $\Phi_{\text{s.t.}} \in \Phi_E$:

1. ass: $f : S \to \mathbb{R}^3$ (intended embedding)
2. $h : S \to \mathbb{R}$ (height function)
3. $\forall i(h(x_{i+1}) > h(x_i))$ (from Φ_{loc})
4. $x_1 \to x_2 \to x_3 \to \cdots \to x_n \to x_1$ (from Φ_{loop})
5. $h(x_2) > h(x_1)$, $h(x_3) > h(x_2)$, $\cdots$, $h(x_n) > h(x_{n-1}), h(x_1) > h(x_n)$ (from 3 and 4)
6. $h(x_1) < h(x_1)$ (from 5 and $\Phi_{\text{s.t.}_3}$)
7. $\neg(h(x_1) < h(x_1))$ (from $\Phi_{\text{s.t.}_1}$)
8. Therefore, $\nexists f$ (from 1, 6 contradicts 7)

As the Penrose steps violate $\Phi_{\text{s.t.}_1}$ or the irreflexivity of $<$ on $\mathbb{R}$, $\Phi_E \cup \{\Phi_{\text{loc}}, \Phi_{\text{loop}}\}$ is unsatisfiable. No embedding $f : S \to \mathbb{R}^3$ exists, such that we can simultaneously satisfy local ascent (Φ_{loc}) and global closure (Φ_{loop}) under a strict total order $<$. At least two possible substitutive soft lifts can help us to overcome the obstruction and fix the structural failure:

(Substitutive soft lift #1): We can drop $\Phi_{\text{s.t.}_1}$ and introduce an additional constraint Φ_{cycle} that permits cyclic orderings. The

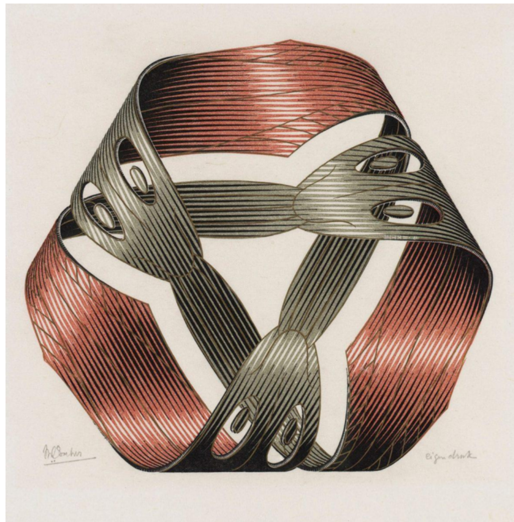

Figure 2: M. C. Escher's (1961) *Möbius Strip I*, wood engraving and woodcut in red, green, gold and black, printed from four blocks. An example of a Möbius-like surface that permits cyclic orderings

> new constraint $(\Phi_E \setminus \Phi_{\text{s.t.}_1}) \cup \Phi_{\text{cycle}}$ allows us to instantiate the Penrose steps on a Möbius-like surface (see Fig. 2 above) where local ascent (Φ_{loc}) is consistent with global closure (Φ_{loop});
>
> (Substitutive soft lift #2): We can drop Φ_{loop} and introduce an additional constraint Φ_∞ that permits an infinitely ascending staircase. Local ascent (Φ_{loc}) is preserved, while we get asymptotic ascent (Φ_∞) instead of global closure (Φ_{loop}).

According to our initial account of structural failure, $\nexists \mathcal{M}(\mathcal{M} \models \Phi)$ (§ 2). When we compare instances of structural failure at φ_1 (the Russell set R and the largest natural number N; see § 3) and instances of structural failure at φ_2 ($x^2 = -1$ and the Penrose steps), it is possible to infer that there exists an inter-level lift $\lambda : \mathcal{M}_{\text{logic}} \uparrow \mathcal{M}_{\text{math}}$ between the former (lower-level) and the latter (higher-level). At least some L_1 (logical) structures

$\mathcal{M}_{\text{logic}}$, including the Russell set R and the largest natural number N, are logically conceivable, syntactically well-formed, yet unsatisfiable under any model. Where soft lifts (subtractive or substitutive) allow us to exclude R and N and retain other logical notions such as the von Neumann universe $\mathcal{V}$ and the set of natural numbers $\mathbb{N}$ for mathematical modelling purposes. By contrast, L_2 (mathematical) structures $\mathcal{M}_{\text{math}}$ are semantic and can be instantiated in at least some models. While R and N collapse under any interpretation, $x^2 = -1$ and the Penrose steps cannot be instantiated only under an intended Δ. Nonetheless, a model of Φ exists in some expanded domain Δ':

$$\nexists \mathcal{M}_\Delta(\mathcal{M}_\Delta \models \Phi) \land \exists \Delta' \exists \mathcal{M}_{\Delta'}((\Delta \subseteq \Delta') \land (\mathcal{M}_{\Delta'} \models \Phi))$$

Whereas the former instances of structural failure are better described as failures arising from logical inconsistency (no model of Φ exists anywhere for R and N), the latter instances are better described as failures arising from obstruction. All things considered, levels L_1 (logic) and L_2 (mathematics) can be characterized in terms of their associated constraints:

(Φ_1/Φ_2^R) Logical conceivability and syntactic well-formedness;
(Φ_2^A) Satisfiability under at least some model;
(Φ_2) Logical conceivability, syntactic well-formedness, and model-theoretic satisfiability.

§5 Structural failure at φ_3

The Vitali (1905) set refers to a set of real numbers that is not Lebesgue-measureable. Δ is the set of real numbers $\mathbb{R}$, $\mathcal{R}$ consists of $\{+, <, \mu\}$ (where

μ denotes the Lebesgue measure), and Φ_L refers to a set of Lebesgue constraints:

(Φ_{count} or countable additivity) $\mu(\bigsqcup_i A_i) = \sum_i \mu(A_i)$ for disjoint, measurable sets;
(Φ_{inv} or translation invariance) $\mu(A + t) = \mu(A)$ for all translations $t \in \mathbb{R}$;
(Φ_{total} or total domain) Every subset of $\mathbb{R}$ is measurable;
(Φ_{AC} or axiom of choice) Given any collection of non-empty sets, it is possible to construct a new set by choosing one element from each set.

Vitali sets can be constructed by following these steps:

1. For any two real numbers x and y in the range $[0, 1]$, group x and y into the same equivalence class whenever $x - y$ is rational. Formally, $x \sim y$ if $x - y \in \mathbb{Q}$.[8]
2. By Φ_{AC}, select exactly one representative from each class. The resulting set may be termed V or the Vitali set.
3. Shift V by every rational number $q \in \mathbb{Q}$ in the range [-1,1]. These shifts $V + q$ are disjoint, cover [0,1] (measure 1), and can be fitted inside [-1,2] (measure 3).

Even though there are uncountably many members in V, we translate V by rational numbers $q \in \mathbb{Q}$ in the range [-1,1]. This yields countably many sets of the form $V + q$ and these sets are disjoint:

[8] This splits [0,1] into exactly one equivalence class containing all rational numbers ($\mathbb{Q}$), since rationals differ from each other by a rational number, and uncountably many other equivalence classes, each consisting of an irrational number – for instance, $\sqrt{2}-1$, $\sqrt{5}-2$, $\frac{\pi}{4}$, and so on – shifted by rationals from some fixed irrational representative.

$[0,1]$ (measure 1) $\subseteq \bigsqcup_q (V+q) \subseteq [-1,2]$ (measure 3)

The constraints Φ_{count} (countable additivity) and Φ_{inv} (translation invariance) force $\mu(V+q) = \mu(V)$ for all q. Therefore:

$1 \leq \sum_q \mu(V) \leq 3$

If $\mu(V) = 0$, then $\sum_q \mu(V) = 0$. If $\mu(V) > 0$, then $\sum_q \mu(V) = +\infty$. However, neither $\sum_q \mu(V) = 0$ nor $\sum_q \mu(V) = +\infty$ lie within the window [1,3]. Given this contradiction, the Vitali set V, though logically conceivable, must also be recognized as a structural failure under the intended interpretation $\mathbb{R}$ with Lebesgue measure μ. At the same time, the structural failure associated with V, a non-measurable subset of [0,1], can be addressed through the following substitutive soft lift:

> Drop Φ_{total} and introduce an additional constraint Φ_σ that restricts the domain of the Lebesgue measure μ to the σ-algebra of Lebesgue-measurable sets. This ensures that V, though visible as sets, are invisible to μ. Therefore, the original obstruction can be overcome with $(\Phi_L \setminus \Phi_{\text{total}}) \cup \Phi_\sigma$, yielding an instantiation of V under $\mathbb{R}$ that remains outside the reach of μ entirely.

The Banach-Tarski paradox, extending the work of Vitali, illustrates how non-measurability in $\mathbb{R}^3$ (3D Euclidean space) becomes so severe that you can even duplicate a solid ball B (Banach and Tarski, 1924) (Fig. 3).

Here, Δ is 3D Euclidean space ($\mathbb{R}^3$), $\mathcal{R}$ is given by $\{A, \mu\}$ (where A denotes rigid motions acting on points in $\mathbb{R}^3$), and Φ_{BT} refers to the set of constraints that gives rise to the Banach-Tarski paradox. Contained within Φ_{BT} are the following constraints:

Figure 3: The Banach-Tarski paradox: given a solid ball B in a 3D Euclidean space $\mathbb{R}^3$, there exists a decomposition of B into a finite number of non-overlapping pieces that can then be put back together to yield two identical copies $B \sqcup B$ of the original ball. Image courtesy of Muljadi (n.d. p. 1)

(Φ_{count} or countable additivity) μ is countably additive;
(Φ_{inv} or invariance: under rigid motions) μ remains invariant under rigid motions such as rotations and transformations;
(Φ_{total} or total domain) μ is defined on all subsets of $\mathbb{R}^3$;
(Φ_{AC} or axiom of choice) Given any collection of non-empty sets, it is possible to construct a new set by choosing one element from each set.

When the Banach-Tarski decomposition of a solid ball B produces the non-measurable pieces $p_1, p_2, \cdots, p_n$, Φ_{AC} licenses the construction of these n pieces. Φ_{inv} permits the reassembly of these n pieces into $B \sqcup B$ by rigid motions alone. Formally:

$B \cong B \sqcup B$, where $\cong$ means 'equidecomposable by rigid motions'

As a result of Φ_{total}, μ must assign a value to every subset of $\mathbb{R}^3$, including the paradoxical, non-measurable pieces $p_1, p_2, \cdots, p_n$. When Φ_{count}

is applied, we get:

$$
\begin{aligned}
V(B) &= \sum_{i=1}^{n} V(p_i) \text{ (from } \Phi_{\text{count}}) \\
V(B) &= V(B \sqcup B) \text{ (from } \Phi_{\text{inv}}, \Phi_{\text{total}}, \text{ and } \Phi_{\text{AC}}) \\
&= \sum_{i=1}^{n} V(p_i) + \sum_{i=1}^{n} V(p_i) \\
&= 2 \times V(B) \text{ (physically impossible)}
\end{aligned}
$$

$x^2 = -1$ and the Penrose steps are instances of structural failure at φ_2 (§ 4), whereas the Vitali set V and the Banach-Tarski paradox are instances of structural failure at φ_3. Again, it is possible to infer the existence of an inter-level lift $\lambda : \mathcal{M}_{\text{math}} \uparrow \mathcal{M}_{\text{phys}}$. $x^2 = -1$, an L_2-level structural failure over $\mathbb{R}$, instantiates a mathematical solution when lifted into $\mathbb{C}$. The Penrose steps, an L_2-level structural failure when embedded under $\mathbb{R}^3$, can be resolved when lifted through additional constraints that permit cyclic orderings (Φ_{cycle}) or infinite ascent ($\Phi_i nfty$). While both the Vitali set V and the Banach-Tarski paradox are perfectly coherent and mathematically instantiable at L_2, their structural failure arises from a failure of compatibility with measure-theoretic constraints such as μ. These measure-theoretical constraints, in turn, are part of a set of nomological constraints that all physical structures $\mathcal{M}_{\text{phys}}$ must satisfy. All things considered, levels L_2 (mathematics) and L_3 (physics) can be characterized in terms of their associated constraints:

(Φ_2/Φ_3^R) Mathematical instantiability and well-formedness, satisfying Φ_2;
(Φ_3^A) Nomological satisfiability or satisfiability in a mathematical model consistent with the laws of physics;
(Φ_3) Nomological and mathematical satisfiability, satisfying both

Φ_2 and Φ_3.

Recall our definition of structure $\mathcal{M}$ in terms of the triple $(\Delta, \mathcal{R}, \Phi)$ (§ 2). At L_3 (physics), Δ consists of a state space $\mathcal{S}$. Observation and measurement (O) allows us to assign values to states $s \in \mathcal{S}$:

> $O : \mathcal{S} \to \mathbb{V}$, where $\mathbb{V}$ denotes the value space appropriate to the physical theory (for instance, $\mathbb{R}$ in classical mechanics, $\mathbb{C}$ in quantum mechanics, and so on)

Furthermore, actual physical processes or dynamics ($\mathcal{F}$) describe how states evolve:

> $\mathcal{F}_t : \mathcal{S} \to \mathcal{S}$ for each $t \in T$, where T denotes the time parameter

O (evaluative) and $\mathcal{F}_t$ (temporal) are relations on $\mathcal{S}$ and can straightforwardly be regarded as a part of $\mathcal{R}$. Last but not least, Φ refers to the physical or nomological constraints that O and $\mathcal{F}$ must satisfy. Physical structure $\mathcal{M}_{\text{phys}}$ can therefore be represented in terms of the triple $(\mathcal{S}, \{O, \mathcal{F}_t\}, \Phi)$:

> (General definition) $\mathcal{M} \coloneqq (\Delta, \mathcal{R}, \Phi)$
> (L_3-specific definition) $\mathcal{M}_{\text{phys}} \coloneqq (\mathcal{S}, \{O, \mathcal{F}_t\}, \Phi)$

We can now better explain why the Vitali set V and the Banach-Tarski paradox are L_3-level structural failures. They succeed under the appropriate L_2-level lift – the removal of the measure-theoretic layer μ – and are mathematically instantiable as $\mathcal{M}_{\text{math}}$. At the same time, our definition $\mathcal{M}_{\text{phys}} \coloneqq (\mathcal{S}, \{O, \mathcal{F}\}, \Phi)$ requires O to be defined and well-behaved on its domain $\mathcal{S}$, since O is the evaluative relation that assigns physical quantities to states. However, as O is undefined with the Vitali set V and the Banach-Tarski paradox, these structures fail to be physically instantiated. The nomological constraints at L_3 can be further specified:

> (Φ_{con} or conservation) The dynamics $\mathcal{F}_t$ preserves the conserved quantities O: $\forall t \in T, \forall s \in \mathcal{S}(\mathcal{F}_t(O(s)) = O(s))$;
> (Φ_{sym} or symmetry) Relations O and $\mathcal{F}_t$ are invariant under a designated symmetry group G: $\forall g \in G, \forall s \in \mathcal{S}(O(g \cdot s) = O(s), \mathcal{F}_t(g \cdot s) = g \cdot \mathcal{F}_t(s))$;[9]
> (Φ_{loc} or locality) O must respect causal structure and measurements at causally disconnected locations cannot influence each other: $(s \sim s') \to [O(s), O(s')] = 0$. This means that measuring $O(s)$ first, then $O(s')$, yields the same outcome statistics as measuring $O(s')$ first, then $O(s)$ (commutativity of observables);
> (Φ_{quant} or quantization) It is impossible, even in principle, to know simultaneously and exactly where a particle is (position) and how fast it is moving (velocity). This is enshrined in the Heisenberg (1927) uncertainty principle: $\sigma_x \sigma_p \geq \frac{\hbar}{2}$, where σ_x denotes the standard deviation of position, σ_p denotes the standard deviation

[9] Symmetry refers to invariance (no change) under transformations (change) or – as the Nobel laureate Frank Wilczek (2016) more succinctly puts it – change without change. Roughly, the group of symmetries acting on mathematical space is the same group acting on physical space. This connection is beautifully supported by Noether's (1918) theorem, according to which continuous symmetries in nature (Φ_{sym}) can be connected to conservation laws and conserved quantities (Φ_{con}). Rotation symmetry is connected to the law of conservation of angular momentum (angular momentum is conserved); translation symmetry is connected to the law of conservation of momentum (momentum is conserved); and time-translation symmetry is connected to the law of conservation of energy (energy is conserved). So compelling and fundamental is the nature of this connection that another Nobel laureate, Steven Weinberg, has seen fit to declare the universe as 'an enormous direct product of representations of symmetry groups' (cited in Gallian, 2002, p. 150).

of momentum, and $\hbar$ denotes the reduced Planck constant;

(Φ_{corr} or correspondence) Relations O and $\mathcal{F}_t$ must recover lower-level physical structures at the limit. For instance:

(a) $\lim_{\substack{v/c\to 0\\ \phi/c^2\to 0}} \mathcal{E}_{\text{rel}} = \mathcal{E}_{\text{Newt}}$, where v denotes the velocity of the system, c denotes the speed of light in a vacuum, ϕ denotes the Newtonian gravitational potential, $\mathcal{E}_{\text{rel}}$ denotes the relativistic equations of motion, and $\mathcal{E}_{\text{Newt}}$ denotes the Newtonian equations of motion;

(b) $\lim_{\hbar\to 0} O_{\text{quantum}} = O_{\text{classical}}$, where $\hbar$ denotes the reduced Planck constant and $\hbar \to 0$ denotes the classical limit of quantum mechanics;

$\vdots$

The inter-level lift $\lambda : \mathcal{M}_{\text{math}} \uparrow \mathcal{M}_{\text{phys}}$ can now be characterized by the nomological constraints that are added at L_3:

$$\Phi_3^A \coloneqq \Phi_{\text{con}} \cap \Phi_{\text{sym}} \cap \Phi_{\text{loc}} \cap \Phi_{\text{quant}} \cap \Phi_{\text{corr}} \cap \cdots$$

Different theories in physics, in turn, can be characterized in terms of a correspondence between maximally satisfiable subsets of Φ_3^A and $(\mathcal{S}, \{O, \mathcal{F}_t\})$. For instance, while dynamics $\mathcal{F}_t$ are typically time-reversal symmetric ($\Phi_{\text{t}-\text{sym}} \subsetneq \Phi_{\text{sym}}$), thermodynamics introduces a temporal asymmetry at the macroscopic level through the second law of thermodynamics ($\Delta S \geq 0$). In a related vein, Φ_{loc} is absent in classical mechanics, since Newtonian space does not have a finite speed limit for signal propagation built into its structure. The formal Newtonian limit, $c \to \infty$, is therefore equivalent to dropping Φ_{loc}. General relativity is characterized by the absence of Φ_{quant}, whereas quantum mechanics QM, given its assumption in favour of quantum entanglement or non-local correlations, drops Φ_{loc}. Quantum field theory QFT can be regarded as a substitutive soft (purely additive) lift, since it drops nothing from QM and merely recovers Φ_{loc}:

$$\Phi_{\text{QFT}} \coloneqq (\Phi_{\text{QM}} \setminus \emptyset) \cup \Phi_{\text{loc}}$$

More generally, each physical theory Θ can be regarded as an intra-level lift ι within L_3, retaining a maximally satisfiable subset of Φ_3^A:

$$\Phi_\Theta \coloneqq \Phi_3^A \setminus \Phi_{\text{dropped}}$$

§6 Conclusion

In conclusion, our theory of structure is based on a methodological wager: we believe that more insight can be gained from studying instances of structural failure than instances of structural success. Under normal conditions, a successfully instantiated structure is overdetermined: the constraints are jointly satisfied and the success-bearing commitments are invisible. When there is structural failure, however, and no model exists to satisfy the full constraint set Φ, the underlying architecture of structure becomes more visible. Our *via negativa* approach – analogous to lesion studies, knockout-gene studies, ablation studies in other disciplines – is grounded in a substantive claim about where insight about structure comes from.

Our formalism makes everything precise. Each structure $\mathcal{M}$ is defined in terms of the triple $(\Delta, \mathcal{R}, \Phi)$ of domains, relations, and constraints. Failure of structure or $\nexists \mathcal{M}(\mathcal{M} \models \Phi)$ is not an absence of structure but rather an informative configuration and object of analysis under our *via negativa* approach. Classical paradoxes and impossibility results (the Russell set R, the largest natural number N, $x^2 = -1$, the Penrose steps, the Vitali set V, and the solid ball B in the Banach-Tarski paradox) are not to be explained away by hand-waving but rather data, encoding the location and nature of obstructions. Through subtractive or substitutive soft lifts, theory and model change point toward how these failed structures can be repaired. Our analysis yields a unified treatment of classical paradoxes and impossibility results, a derived rather than stipulated level hierarchy, emergence and reduction as structural properties that can be read off the lifts, and an account of model and theory change in terms of soft lifts and subject to the demands of satisfiability. We note that our analysis has been deliberately confined to logic (L_1), mathematics (L_2), and physics (L_3). We anticipate, as a future and genuine research programme, upward extensions: chemical constraints (φ_4) at L_4, biological constraints (φ_5) at L_5, and potentially further levels beyond. If our framework is generalizable, then we should expect each of these levels to exhibit its own characteristic structural failures, analogous to the paradoxes, impossibility results, and soft lifts that we have identified in L_1–L_3. Furthermore, our *via negativa* approach implies that these structural failures will continue to function as the most fruitful sites of inquiry.

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